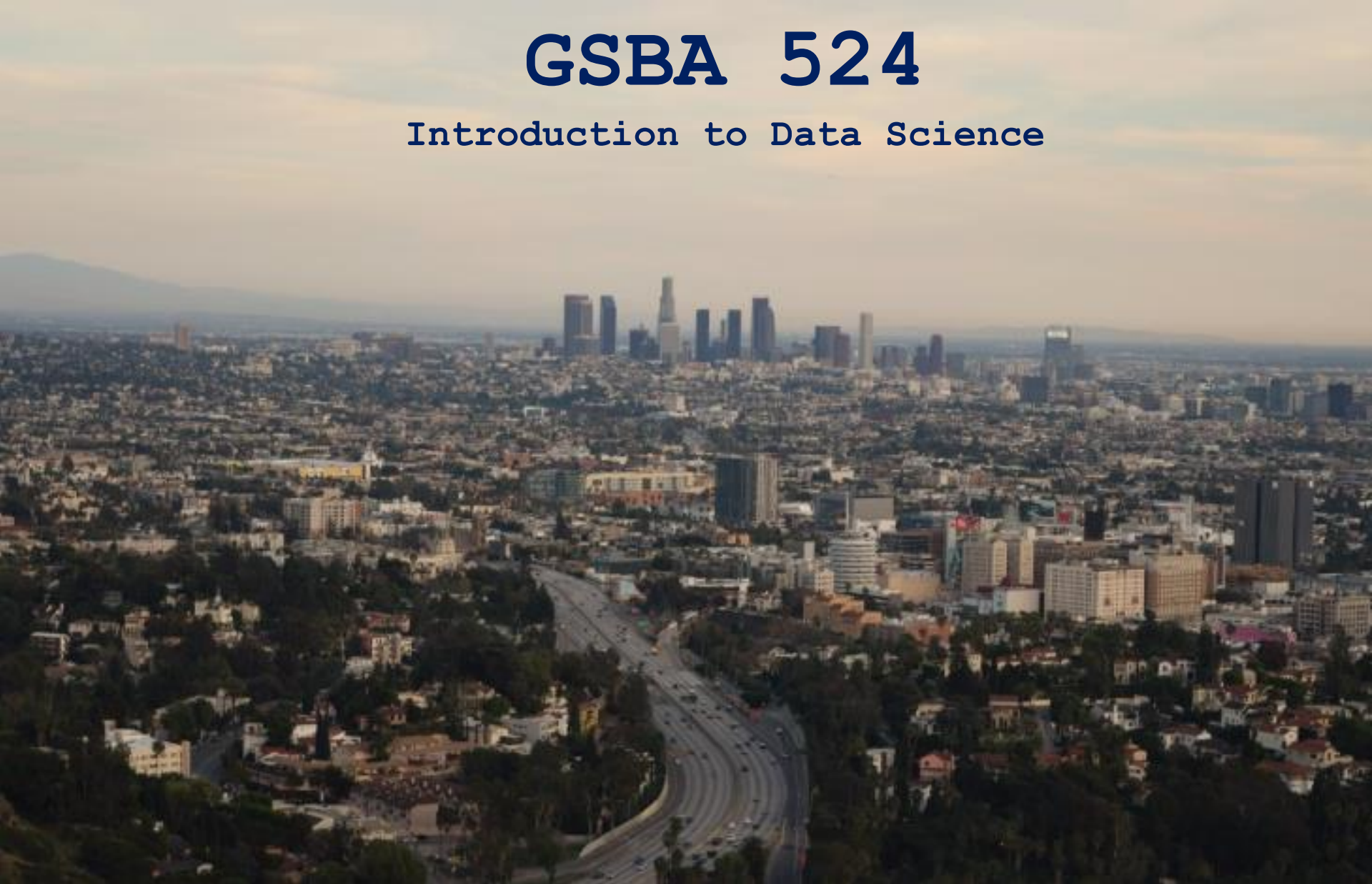


GSBA 524

Introduction to Data Science



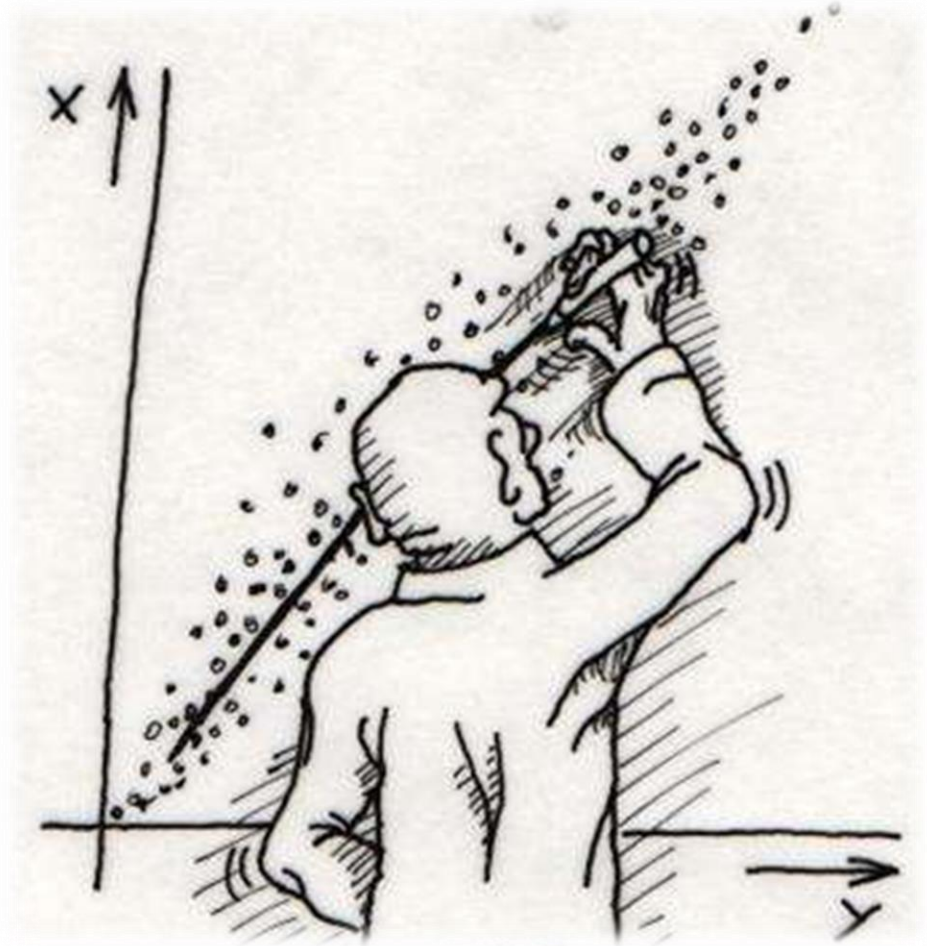
G. Mukherjee

Lecture #11

OUTLINE

Simple Linear Regression

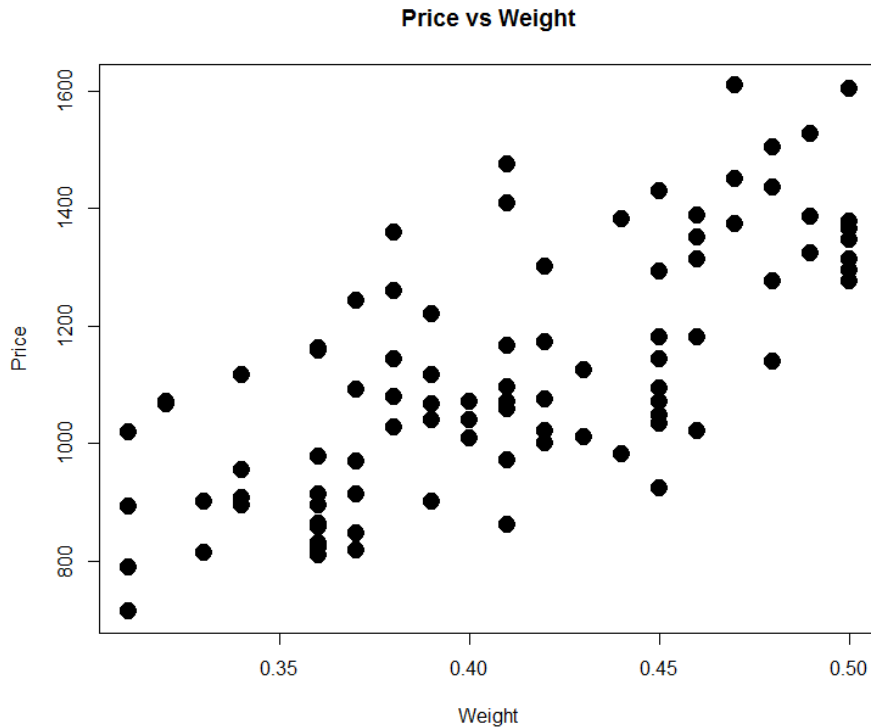
- ❑ Linear Regression Model
- ❑ Intercept & Slope estimation



Simple Linear Regression Model

- A model for predicting 1 response variable (Y) based on one explanatory variable (X)
- A **response (dependent)** variable is the variable of interest (measures the outcome of a study)
- An **explanatory (independent)** variable explains or causes changes in the response variable

Motivating Data: Diamonds 😊



- Data: 93 diamonds of similar clarity
- Response variable: Price (in \$)
- Explanatory Variable: Weight (in Carats)

Example: two questions

- What's the average price of diamonds that weigh 0.4 carat?
- How much more do diamonds that weigh 0.45 carat cost?

Straight lines

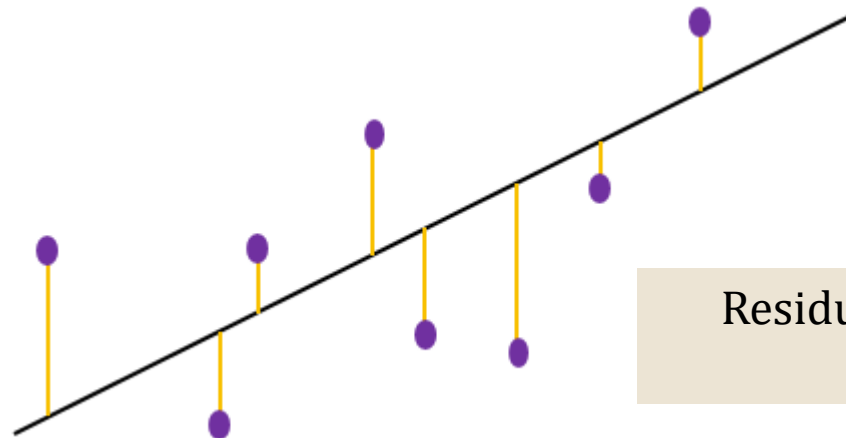
Suppose that Y is a response variable, X is an explanatory variable. A straight line relating Y to X has an equation of the form

$$y = b_0 + b_1 x$$

Here b_1 is called the **slope**, the amount by which y changes when x changes by one unit. The term b_0 is the **intercept**, the value of y when $x = 0$.

Regression and least squares

Imagine fitting a line through some data



Residual = actual-predicted
 $\hat{e}_i = y_i - \hat{y}_i$

Notations

y_i : actual response value of the i th observation

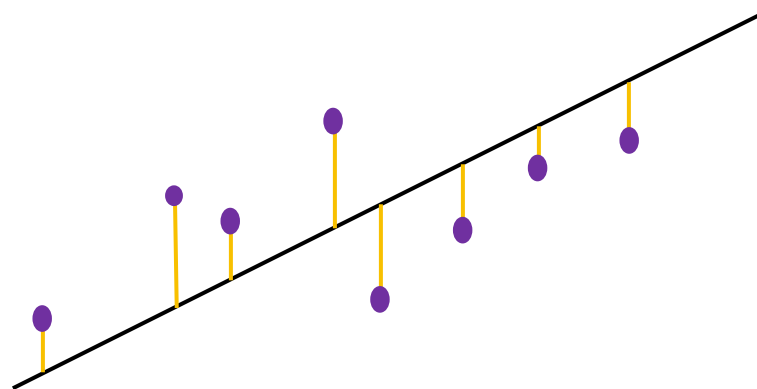
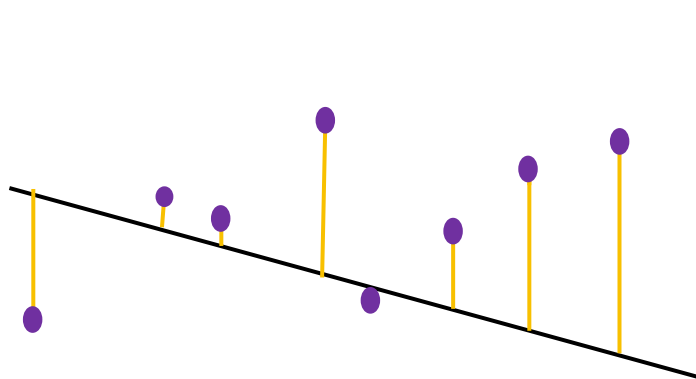
$\hat{y}_i$: Predicted/fitted value for i th observation

$\hat{e}_i$: residuals for the i th observations

The least squares criterion

Want residuals to be small:

Minimize the sum of squared residuals



Residual = actual-predicted
 $\hat{e}_i = y_i - \hat{y}_i$

bad fit

better fit

Least-squares regression line

The line that makes the sum of squared residuals as small as possible is called the Least-squares line.

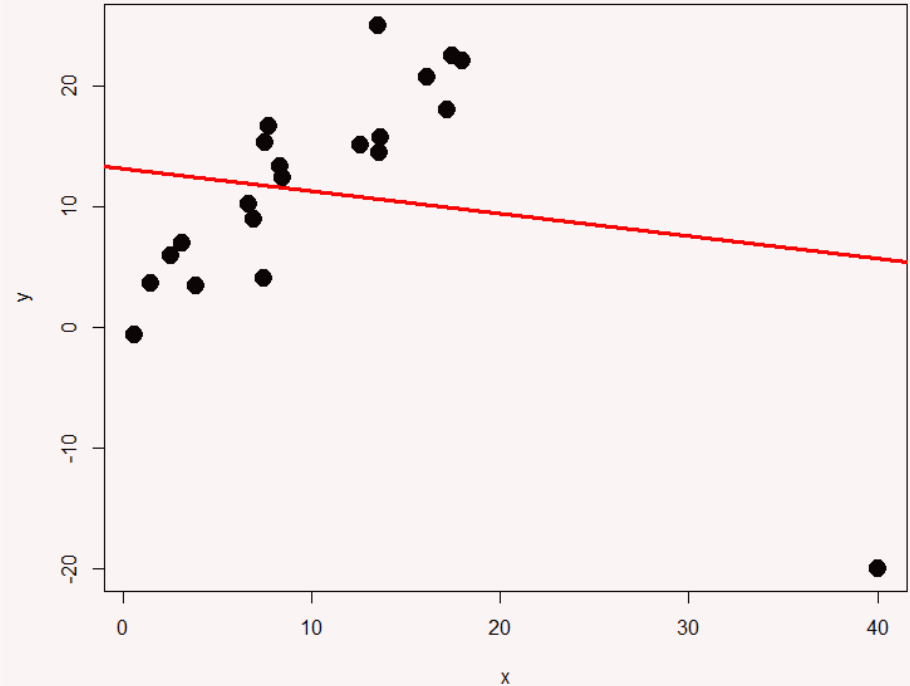
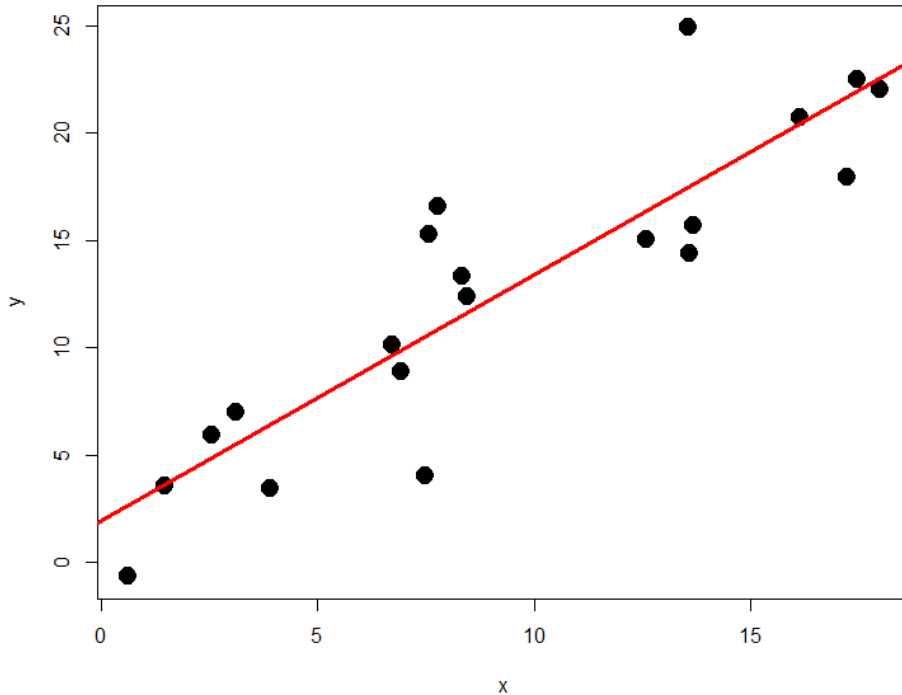
The equation of the least-squares regression line of y on x is

$$\hat{y} = b_0 + b_1x$$

with **slope** $b_1 = r \frac{s_y}{s_x}$ and **intercept** $b_0 = \bar{y} - b_1\bar{x}$

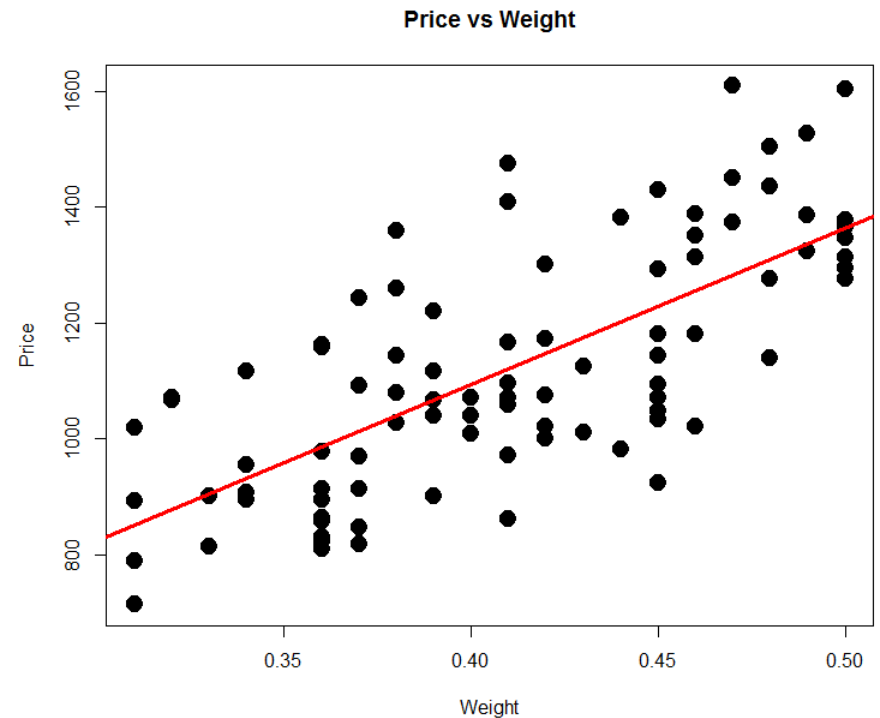
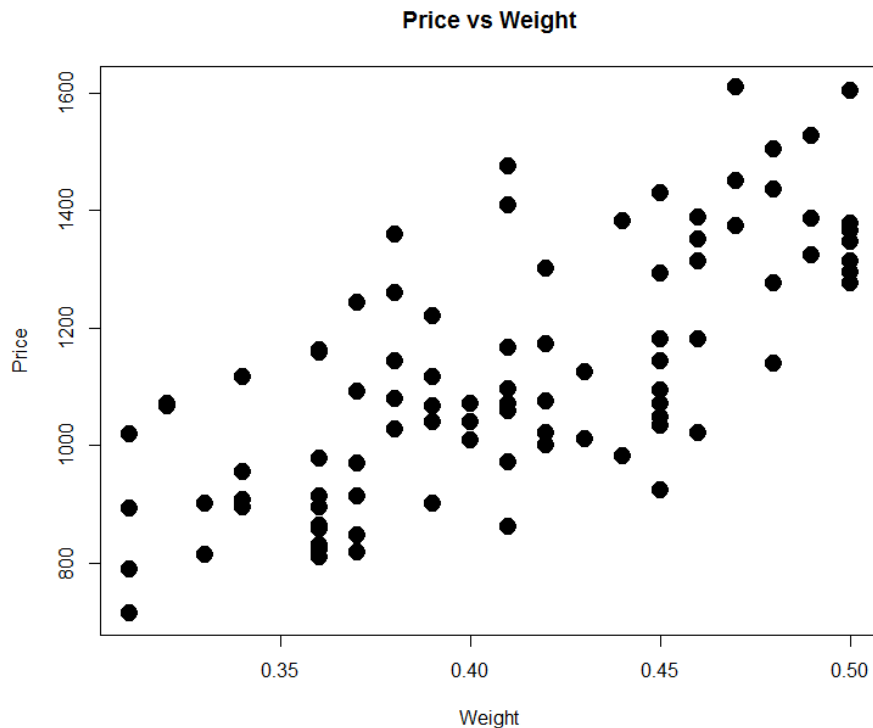
r is correlation between Y and X, s_y is SD of Y, s_x is SD of X
 $y\text{-bar}$ is mean of Y, $x\text{-bar}$ is mean of X

Least-squares regression is not robust



Just adding an outlier can swing the least squares line by lot ☹

Back to the diamond example



```
Call:
lm(formula = Price ~ weight, data = data)

Coefficients:
(Intercept)      weight 
      15.2         2697.3
```

$$\text{Estimated Price} = 15.2 + 2697.3 \text{ Weight}$$

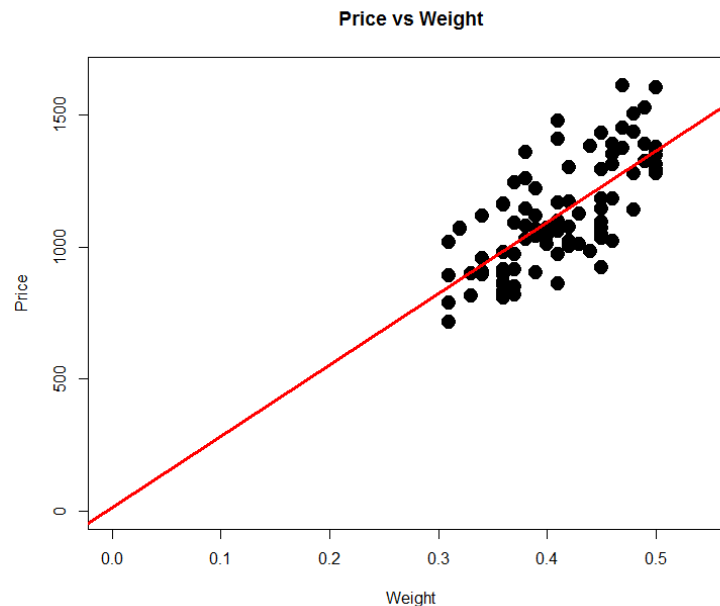
Predicting from the Fitted Line

The average price of a diamond that weighs 0.4 carat is

$$\textit{Estimated Price} = 15.2 + 2697.3 * 0.4 = \underline{\underline{\$1,094.12}}$$

Interpreting the Intercept

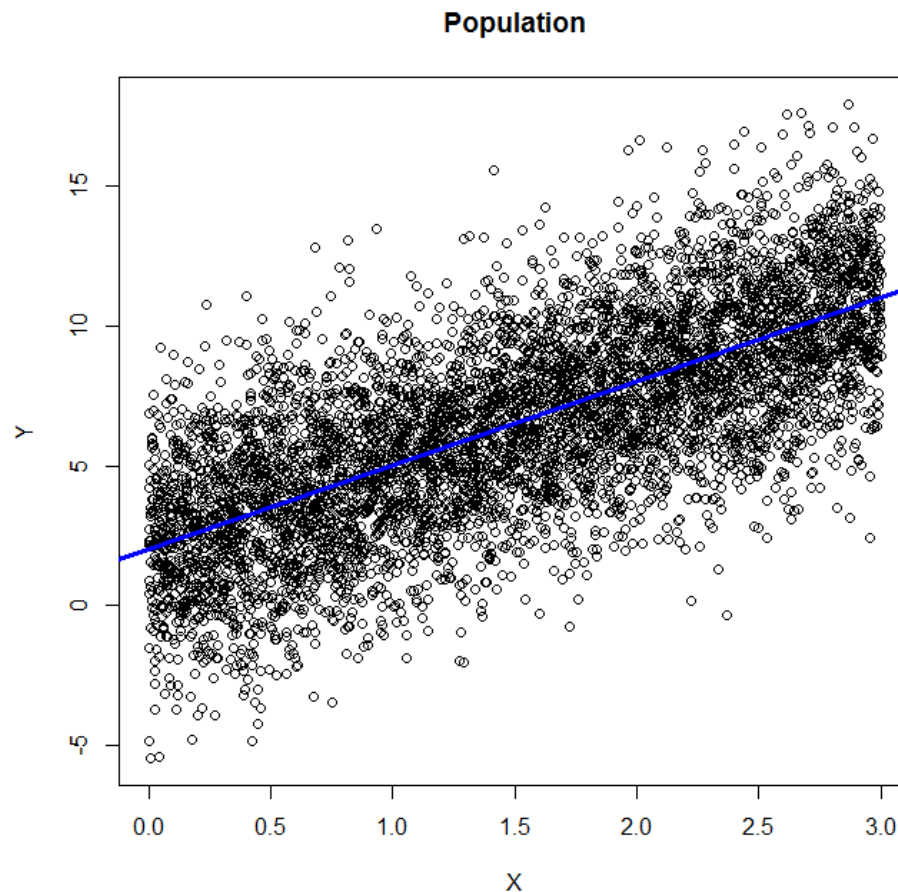
- The intercept is the portion of y that is present for all values of x (i.e., fixed cost, \$15.2, per diamond).
- The intercept estimates the average response when $x = 0$ (where the line crosses the y axis).
- *Unless the range of x values includes zero, b_0 will be an extrapolation.*



Interpreting the Slope

- ❖ The slope describes how differences in the explanatory variable associate with differences in the response
(units of the slope are units of Y divided by units of X)
- ❖ In our example the slope is \$2,697.3 per carat.
- ❖ While tempting, it is not correct to describe the slope as *the change in Y caused by changing X*.

Regression line changes from sample to sample



Population (Census Study)

True linear relation is
 $y = \beta_0 + \beta_1 x = 2 + 3x$

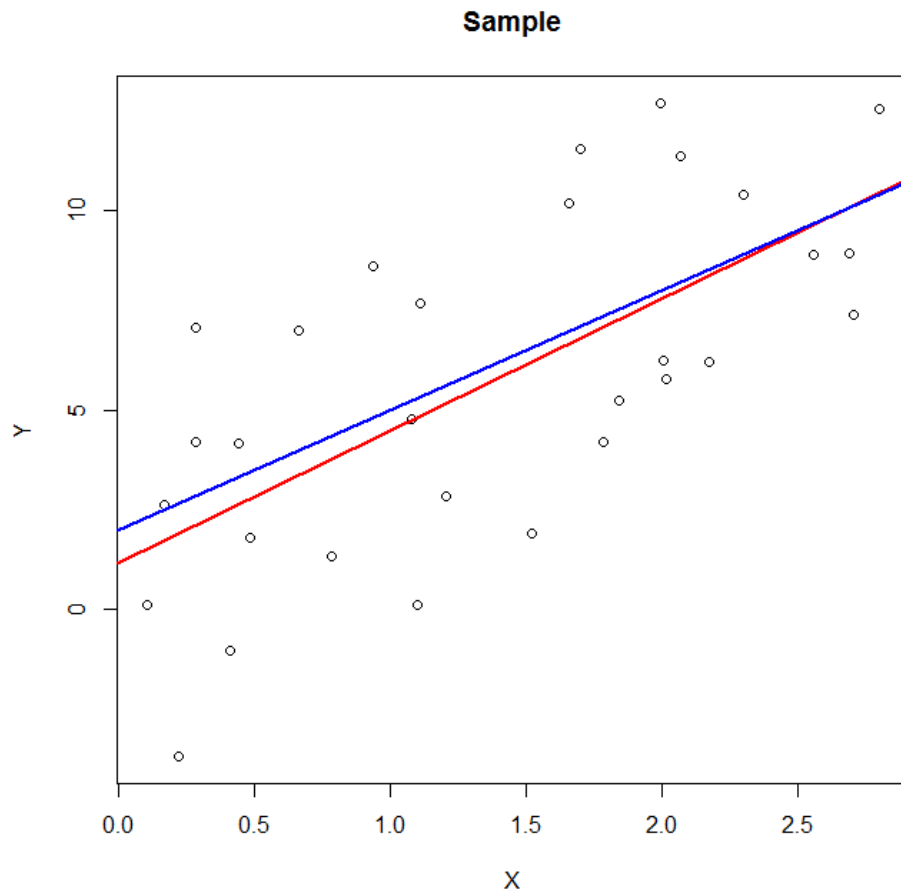
Example: population vs sample

Sample relation based on LS line

$$b_0 + b_1x = 1.2 + 3.3x$$

Population line

$$\beta_0 + \beta_1x = 2 + 3x$$



- If we get a random sample from the population the Least square line gives us an estimate of the population coefficients based on the sample.
- If the sample changes the estimates will change.
- If sample size is good enough, the estimates will be close to the population.

Interpreting the R output

```
> summary(model)
```

call:

```
lm(formula = Price ~ Weight, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-305.96	-92.21	-24.07	95.06	354.93

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	15.2	114.8	0.132	0.895
weight	2697.2	277.9	9.705	1.06e-15 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 144.9 on 91 degrees of freedom

Multiple R-squared: 0.5086, Adjusted R-squared: 0.5032

F-statistic: 94.18 on 1 and 91 DF, p-value: 1.058e-15



GitHub
Copilot

Coefficient
Table

The copilot prompt for this is:

```
#show the summary of the model
```

T tests in the coefficient table

Testing
For
Intercept

To test the hypothesis $H_0: \beta_0 = 0$ vs $H_A: \beta_0 \neq 0$
compute the test statistic

$$t = \frac{b_0}{se(b_0)}$$

then use $t(n-2)$ distribution to find the p-value (two-tail)

Testing
For
Slope

To test the hypothesis $H_0: \beta_1 = 0$ vs $H_A: \beta_1 \neq 0$
compute the test statistic

$$t = \frac{b_1}{se(b_1)}$$

then use $t(n-2)$ distribution to find the p-value (two-tail)

Coefficient table

Predictor	Coef	SE Coef	T	P
Intercept	15.2	114.8	0.132	0.895
Weight	2697.2	277.9	9.705	0.000

Predictor	Coef	SE Coef	T	P
Intercept	b_0	$se(b_0)$	$b_0/se(b_0)$	$p\text{-value}$
X variable	b_1	$se(b_1)$	$b_1/se(b_1)$	$p\text{-value}$

p-value:

corresponds to the T-statistic;

is the two-tail probability under the t-distribution with $n-2$ df;

used for testing $H_0: \beta_1 = 0$ vs. $H_A: \beta_1 \neq 0$

Interpretation of the hypothesis test for Slope

Suppose we can reject $H_0: \beta_1 = 0$ at significance level α .

This result can be interpreted as follows:

At level α ,

- β_1 is statistically significantly different from zero.
- Linear association between X and Y is statistically significant.
- X is a statistically significant predictor of Y.